

SmartMeds (Medicine Dispenser)

Literature Survey — 2025

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1. Introduction

This literature survey reviews recent academic and technical developments (2018–2025) related to smart medication dispensers, IoT pill boxes, voice-activated medical systems, medication adherence devices, and speech recognition integration with Raspberry Pi-based dispensers.

2. Objectives of the Survey

- Identify state-of-the-art hardware and software designs for smart medication dispensers.
- Explore voice-controlled and voice-assisted technologies.
- Summarize usability and adherence evidence.
- Provide direct links for reference and citation.

3. Methodology

Searches were conducted on PubMed, Springer, MDPI, ScienceDirect, and ResearchGate using keywords like “smart medication dispenser,” “IoT pill box,” and “voice medicine dispenser.” Priority was given to open-access and recent studies.

4. Selected Papers (Representative Summaries & Links)

1. Gargioni L. et al. — “A Systematic Review on Pill and Medication Dispensers” (2024)

Comprehensive review covering dispenser designs and evidence gaps.

Link: <https://pubmed.ncbi.nlm.nih.gov/PMC11052969>

2. Patel T. et al. — “Usability of an Automated Medication Dispensation Device and Adherence Dashboard: A Study Protocol” (2024)

Describes usability evaluation frameworks with SUS and NASA-TLX metrics.

Link: <https://pubmed.ncbi.nlm.nih.gov/PMC11575765>

3. Nasir Z. et al. — “Design of a Smart Medical Box for Automatic Pill Dispensing and Health Monitoring” (2023)

Raspberry Pi-based smart box with biometric and vital sensors.

Link: <https://doi.org/10.3390/engproc2023032007>

4. Dayananda P., Upadhya A. — “Development of Smart Pill Expert System Based on IoT” (2024)

SPEC 2.0 IoT-enabled pill system with mobile control and anti-overdose protection.

Link: <https://link.springer.com>

5. Sree VB. — “Smart Medicine Pill Box Reminder with Voice and Display for Home Use” (2020)

Prototype integrating voice reminders with visual notifications.

Link: <https://www.sciencedirect.com>

6. Peddisetti V. et al. — “Enhancing Medication Adherence with an IoT-Based Pill” (2024)

Describes BLE-based cup verification system for adherence.

Link: <https://www.fullerton.edu>

7. Olaleye SA. — “Unveiling the Complementariness of Robotic Tablet Dispensers” (2024)

Bibliometric analysis identifying automation research gaps.

Link: <https://www.sciencedirect.com>

8. Ghazal M. — “Deep Prescription Understanding and Low-Cost Medication Dispenser” (2025)

AI-driven prescription parsing and dispensing schedule mapping.

Link: <https://www.sciencedirect.com>

9. ResearchGate — “Real-Time Use of Automatic Voice Command Drug Dispenser (VCMD)”

Prototype demonstrating voice-commanded dispenser usability.

Link: <https://www.researchgate.net>

10. Faisal S. et al. — “Key Features of Smart Medication Adherence Products” (2023)

Comprehensive review outlining essential SMAP design features.

Link: <https://aging.jmir.org>

11. Abhijith M. et al. — “IoT-Based Smart Medicine Dispenser: A Technological Solution for Medication” (2025)

ESP32-based dispenser enabling automated dispensing and adherence tracking.

Link: <https://doi.org/10.52783/jns.v14.3575>

12. Chanuka Bandara et al. — “Automated Medicinal-Pill Dispenser with Wireless and Cellular Connectivity” (2023)

Wireless and cellular pill dispenser with web interface.

Link: <https://www.astesj.com/v08/i03/p18/>

13. Pang W. L. et al. — “Smart IoT Mobile Medication Dispenser” (2023)

Low-cost IoT dispenser controlled via mobile app.

Link: <https://www.ijisae.org/index.php/IJISAE/article/view/3078>

14. Manipal Academy — “IoT and ML-based Smart Pill Dispenser (SPD) for Elderly Monitoring” (2023)

Combines IoT and ML for elderly medication adherence tracking.

Link: <https://doi.org/10.1109/iSES58672.2023.00094>

15. Ali M. et al. — “Design of a Smart Medical Box for Automatic Pill Dispensing and Health Monitoring” (2023)

Low-cost dispenser using sensors and SMS alerts.

Link: <https://www.mdpi.com/2673-4591/32/1/7>

5. Gaps Identified & Opportunities for SmartMeds

- Voice interaction lacking; offline ASR offers improvements.
- Ingestion verification remains underexplored.
- Accessibility for seniors and visually impaired needs focus.
- Security (biometric/anti-tampering) is underused.
- Limited clinical trials on adherence outcomes.